MIL-S-55488B
16 August 1976
SUPERSEDING
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5 January 1973

MILITARY SPECIFICATION

SIGHT, NIGHT VISION,
INDIVIDUAL SERVED WEAPONS,
AN/PVS-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a night vision sight for individual served weapons, AN/PVS-2.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Packaging Methods of
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam Expanded Bead)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MIL-P-19834	Plates, Identification, or Instructions, Metal Foil, Adhesive Backed, General Specification for
MIL-P-26514	Polyurethane Foam, Rigid or Flexible for Packaging

STANDARDS

FEDERAL

FED-STD-356	Commercial Packaging of Supplies and Equipment

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallets, Skids, Runners, or Pallet Type Base
MIL-STD-171	Finishing of Metal and Wood Surfaces
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-645710	Night Vision Sight, Individual Served Weapons

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Description. The night vision sight, individual served weapons herein after called viewer, shall consist of a housing, spare parts, manual, and a carrying case, boresight assembly, high voltage oscillator (when applicable), and battery.

3.2 Construction. The viewer shall be constructed in accordance with SC-DL-645710 and as specified herein.

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3.3 First article. The supplier shall furnish and test assemblies as specified in the contract or purchase order (see 6.2).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on the applicable drawings. Parts and materials not specified shall be subject to all provisions of this specification and shall be in accordance with MIL-P-11268.

3.5 Operation and environmental.

3.5.1 Mechanical shock. The viewer shall not be damaged (see 6.3) nor shall boresight alignment change more than one half milliradian when mounted on the M-14 rifle and 1000 rounds (First Article Inspection) or 500 rounds (quality conformance inspection) are fired from the rifle.

3.5.2 Vibration. The viewer shall not be damaged during and after vibration at a frequency of 300 hertz (Hz) at an amplitude of not less than 0.0625 inch (total displacement of 0.1250 inch) for periods of 5 minutes each along each of the following axis:

a. Parallel to the optical axis.

b. Perpendicular to the optical axis in the vertical plane.

3.5.3 High and low temperature storage, operation and thermal shock. The viewer shall not be damaged by high and low temperature storage, operation and thermal shock under the following conditions:

a. Storage	At plus 68°C for 4 hours and at minus 54°C for 12 hours with humidity controlled at 90 to 98 percent at and above room temperature (see 6.3) and uncontrolled humidity below room temperature.
b. Operation	At temperatures from plus 52°C to minus 54°C at maximum humidities possible throughout the temperature range.
c. Thermal Shock	From room temperature to plus 68°C, from plus 52°C to room temperature; from room temperature to minus 54°C and from minus 54°C to room temperature.

3.5.4 Humidity. The viewer shall not be damaged when exposed to humidity as follows:

From room temperature to plus 52°C, and from plus 52°C to room temperature at 90 to 98 percent relative humidity.

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3.5.5 Salt fog. The viewer shall operate as specified herein and shall not be damaged after having been exposed to salt fog in accordance with Method 509 of MIL-STD-810.

3.5.6 Immersion. The viewer shall be watertight, shall operate as specified herein, and shall not be damaged after immersion to a depth of 3 feet for 30 minutes in fresh water.

3.5.7 Modulation transfer-objective lens. The modulation transfer (see 6.3) in the spectral range of an S-20 photoemissive surface to a tungsten source operating at a color temperature of 2870° Kelvin (K) shall be equal to or greater than the values for the lens assembly listed in Table I.

TABLE I Modulation transfer objective

RESOLUTION	MODULATION TRANSFER IN PERCENT	
LP/MM on the Photocathode	Measured on Optical Axis	3.5° off Optical Axis Measured Tangentially
4.0	90	89
7.0	85	82
16.0	76	73

All lenses shall be essentially free from aberration. Upon rotation of any lens or mirror 90° in a plane perpendicular to the optical axis of the objective, the modulation transfer value shall not degrade more than 5 percent from the on-axis measurements.

3.5.8 Modulation transfer-eyepiece. The modulation transfer at zero diopter setting for the region of spectral response to a P-20 phosphor shall be equal to or greater than the value listed in Table II.

TABLE II Modulation transfer-eyepiece

RESOLUTION	MODULATION TRANSFER VALUES IN PERCENT
LP/MM on the Photocathode	
2.5	97
7.0	92
10	88
20	77

All lenses shall be essentially free from aberration. Upon rotation of any lens 90° in a plane perpendicular to the optical axis of the eyepiece, the modulation transfer shall not degrade more than 5 percent.

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3.5.9 Field of view. The field of view of the viewers shall not be less than 180 milliradians.

3.5.10 Stray light luminance. Stray light luminance contributions through the systems over the image of a disk illuminated at 5 x 10^{-3} foot-lambert subtending a solid angle of 10.5°, plus or minus 0.5° with a white surrounding surface illuminated at 5 x 10^{-3} foot-lambert subtending a solid angle of 120° minimum shall not exceed 3 percent of the luminance of the illuminated disk measured through the system without the white surrounding surface illuminated.

3.5.11 Focus adjustment.

3.5.11.1 Objective lens. Objective lens focus adjustment shall insure viewing of sharp clear images of objects located at distances from 4 meters to infinity. Objective torque shall be not less than 8, nor greater than 26 inch-pounds.

3.5.11.2 Eyepiece lens. Eyepiece lens foucs adjustment shall insure eyepiece focus of plus 4, to minus 4 diopters by applying not more than 7.0 inch-pounds of torque with an indexing error not exceeding plus or minus 0.25 diopter.

3.5.12 Image roll. Focusing the objective through the range specified in 3.5.11 shall cause not more than one milliradian of apparent rotation of the scene image relative to the reticle image.

3.5.13 Boresight adjustment. The viewer shall be adjustable over not less than 3° in both elevation and azimuth. Adjustment in azimuth shall not change elevation, nor shall adjustment of elevation change the azimuth. The reticle patterns shall be in clear sharp focus. There shall be no parallax between the reticle and image.

3.5.14 Click movement. Adjustment of boresight shall be indexed by means of audible clicks. Each click shall move the reticle pattern 0.5 milliradian with respect to its mount. Boresight knob torque shall not exceed 8 inch-pounds.

3.6 Electromagnetic compatibility. The electromagnetic compatibility of the viewer shall be in accordance with MIL-STD-461, Table A-1-MIL-E-55301, Equipment Class; Tactical Support, NON-CE.

3.7 Identification markings. Identification markings shall be in accordance with the applicable drawings and MIL-STD-130. Plates shall conform to MIL-P-19834.

3.8 Treatment and painting. The portions of the PVS-2 and its components and parts normally painted shall be cleaned, treated and painted in accordance with the applicable part drawing and in accordance with MIL-STD-171.

3.9 Government loaned property. The following property in the quantities indicated will be loaned by the Government (see 6.4).

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Item No.	Description	National Stock No.	Quantity
1	Rifle, 7.62 MM M-14 W/E	1005-589-1271	6
2	Cartridge 7.62 MM NATO	1305-301-1676	As required See 6.4
3	Intensifier Assembly, 25 millimeter	5855-087-2948 or 5855-177-3502	As required See 6.2

3.10 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwsie specified by the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirement specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Parts and materials inspection. The supplier is responsible for insuring that parts and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as specified herein.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.5).

c. Inspection of preparation for delivery (see 4.9).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 6.2).

4.3.1 Inspection. Each viewer shall be inspected for defects in accordance with Table III. One or more defects shall be cause for rejection.

TABLE III Inspection

Defects	Requirement paragraph
MAJOR	
101. Boresight adjustment knob binds.	3.5.13
102. Objective binds while focusing.	3.5.11.1
103. Eyepiece diopter adjustment binds.	3.5.11.2
104. External hardware on viewer loose.	3.2
105. Intensifier does not energize when viewer is turned ON.	3.4

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TABLE III Inspection (con't)

Defects	Requirement paragraph
MINOR	
201. Finish on viewer and carrying case does not conform to MIL-STD-171 and applicable drawings.	3.8
202. Identification markings and plates not in accordance with applicable drawings.	3.7
203. Workmanship not as specified.	3.10

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the first article viewer shall be subjected to the tests in Table IV. Tests shall be conducted in the order listed. Failure of any test shall be cause for rejection.

TABLE IV First article inspection

Inspection	Requirement paragraph	Test paragraph
Modulation transfer-objective	3.5.7	4.7.1
Modulation transfer-eyepiece	3.5.8	4.7.2
Mechanical shock	3.5.1	4.6.1
Vibration	3.5.2	4.6.2
Humidity	3.5.4	4.6.4
High and low temperature operation and thermal shock	3.5.3	4.6.3
Immersion	3.5.6	4.6.6
Modulation transfer-objective	3.5.7	4.7.1
Modulation transfer-eyepiece	3.5.8	4.7.2
Field of view	3.5.9	4.7.3
Focus adjustment	3.5.11	4.7.5
Image roll	3.5.12	4.7.6
Boresight adjustment	3.5.13	4.7.7
Click movement	3.5.14	4.7.8
Stray light luminance	3.5.10	4.7.4
Electromagnetic compatibility	3.6	4.8
Salt fog	3.5.5	4.6.5

4.4 Inspections covered by subsidiary documents. The viewer shall be inspected under the applicable subsidiary doucments as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order (see 4.3.1 and Table III).

4.5 Quality conformance inspection.

4.5.1 Examination. Each viewer shall be examined as specified in 4.3.1. Presence of one or more defects shall be cause for rejection. Examination will be performed both before and after the successful completion of testing (Group A, B and C as applicable).

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4.5.2 Group A inspection. Each viewer shall be inspected in accordance with Table V. Failure of any test shall be cause for rejection.

TABLE V Group A inspection

Inspection	Requirement paragraph	Test paragraph
Modulation transfer-objective	3.5.7	4.7.1
Modulation transfer-eyepiece	3.5.8	4.7.2

4.5.3 Group B inspection. Group B inspection shall be conducted on viewers selected from units which have passed the examination and tests in 4.5.1 and 4.5.2. Sampling shall be in accordance with MIL-STD-105, AQL 2.5 percent defective, inspection level S-3. Tests shall be conducted in accordance with Table VI and may be performed in any sequence except that the modulation transfer tests shall be last.

TABLE VI Group B inspection

Inspection	Requirement paragraph	Test paragraph
Immersion	3.5.6	4.6.6
Vibration	3.5.2	4.6.2
Modulation transfer-objective	3.5.7	4.7.1
Modulation transfer-eyepiece	3.5.8	4.7.2

4.5.4 Group C inspection. This inspection shall consist of the tests specified in Table VII and shall be performed on sample units that have been subjected to and met Group A inspection. Sample units shall be selected in accordance with 4.5.4.1. The tests may be performed in any order except that the modulation transfer tests shall be last.

4.5.4.1 Sampling for inspection. Three sample viewers shall be selected monthly by the Government for all inspections in Table VII whenever the monthly quantity specified in the contract starts quality conformance testing or every 30 days whichever occurs first. The three units are not associated with inspection lots as prescribed by MIL-STD-105. The only selection prerequisite is that the three units conform to normal production processing. The first three Group C units shall begin test at the time the first quality conformance lot (see 4.5.3) begins Group B inspection.

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TABLE VII Group C inspection

Inspection	Requirement paragraph	Test paragraph
Mechanical shock	3.5.1	4.6.1
Humidity	3.5.4	4.6.4
High and low temperature operation and thermal shock	3.5.3	4.6.3
Salt Fog	3.5.5	4.6.5
Field of view	3.5.9	4.7.3
Stray light	3.5.10	4.7.4
Focus adjustment	3.5.11	4.7.5
Image roll	3.5.12	4.7.6
Boresight adjustment	3.5.13	4.7.7
Click movement	3.5.14	4.7.8
Modulation transfer-objective	3.5.7	4.7.1
Modulation transfer-eyepiece	3.5.8	4.7.2

4.5.4.2 Group C failures. Actions required related to Group C failures shall be as specified in the contract, (see 6.2).

4.5.5 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Test methods. Tests required to determine compliance with this specification shall be conducted in accordance with the test procedures specified herein. All test gages and instruments used during tests shall have been calibrated within 30 days prior to the start of testing and thereafter as required, or at intervals not to exceed 6 months. Unless otherwise specified all tests shall be performed at room temperature (see 6.3). Test facilities, chambers, and apparatus used in conducting tests shall meet the requirements of MIL-STD-810.

4.6.1 Mechanical shock. The test may be conducted during daylight hours. The number of rounds to be fired shall be as follows:

	M-14 Rifle
a. For first article testing	1000 rounds
b. For quality conformance testing	500 rounds

The viewer shall be mounted on the M-14 rifle by means of a weapon adapter bracket. After mounting the viewer, fire 20 rounds of ammunition through the rifle. After 20 rounds have been fired, the rifle, with the viewer attached shall be mounted on a rigid holding device with the viewer focused and

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sighted on a target 20 to 30 feet away. Adjust the viewer boresight until the reticle coincides with the center of the target. The target shall have a clearly marked grid pattern. Each square on the grid shall represent a movement equal to 0.5 milliradian. All hardware and connectors shall be checked to insure that no parts are loose. The weapon and viewer shall be removed from the device and 200 rounds shall be fired from either the prone position or the off-hand position. During and after the firing of 200 rounds, observe and record any internal flashing, flickering, or deterioration in quality of the image presented by the viewer; or any loosening or failure of any hardware or adjustment. After 200 rounds are fired, the weapon and viewer shall be replaced in the holding device and the position of the reticle with respect to the original settings shall be determined. Any change in boresight or reticle shift, or any evidence of failure to conform to the requirements of this specification shall constitute failure of test. The 200 rounds firing and examination cycle shall be repeated as often as required to accumulate the total number of rounds specified to be fired. No adjustment, tightening of parts, or repairs of any kind shall be permitted after the initial boresighting is completed and the test firing cycles have begun without proper recording of the same. Should any internal flashing, flickering or deterioration in quality of the image presented by the viewer or any loosening or failure of any hardware or adjustment occur during or after any firing cycle, the test has failed and no further testing to complete that test will be performed.

4.6.2 Vibration. Mount the viewer rigidly to the vibration table. Subject the viewer to a simple harmonic motion parallel to the optical axis and perpendicular to the optical axis in the verticle plane at an amplitude of 0.0625 inch (0.1250 inch total excursion) at a frequency of 30 Hz for 5 minutes in each position. Examine the viewer. Evidence of any shock and vibration damage shall constitute failure of this test.

4.6.3 High and low temperature storage, operation and thermal shock. Place the viewer in a chamber which has an internal temperature of plus 68°C and soak for 4 hours. Lower the temperature in the chamber to plus 52°C in not more than 2 hours and soak for 4 hours. Turn the viewer ON and observe for image intensifier operation for the last 30 minutes of the twelfth hour. It is permissible to replace the battery in the unit at this time, with a battery which has been at room temperature in order to operate this unit. Remove the viewer from the chamber and soak for 4 hours at room temperature. Turn the viewer ON and observe for image intensifier operation for the last 10 minutes of this period. This constitutes one complete 28 hour cycle as shown in Figure 1. Repeat the above cycle. Failure to operate or evidence of any operation and storage damage during or immediately following the above two cycles shall constitute failure of this test.

4.6.4 Humidity. Place the viewer in a chamber at room temperature and a relative humidity of 90 to 98 percent and soak for 4 hours. Raise the temperature of the chamber to plus 52°C in not more than 2 hours, maintain the relative humidity of 90 to 98 percent and soak for 4 hours. Lower the temperature of the chamber to room temperature in not more than 2 hours. This constitutes one complete cycle as shown in Figure 2. Perform the above cycle

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6 times. Turn the viewer ON and observe for image intensifier operation for the last 30 minutes of the room temperature portion of the 4th and 6th cycles. Failure to operate or evidence of any operation and storage damage during, or immediately following, the above 6 cycles shall constitute failure of this test.

4.6.5 Salt fog. Test the viewer in accordance with Method 509 of MIL-STD-810 except that a 20 percent salt solution shall be used. Failure to meet the requirements of MIL-STD-810 or any evidence of damage shall constitue failure of this test.

4.6.6 Immersion. Immerse the viewer to a minimum depth of 3 feet in fresh water for a period of 30 minutes. Dry all external surfaces, disassemble and inspect for entrance of water or moisture. Any evidence of water or moisture within the viewer shall constitute failure of this test.

4.7 Performance.

4.7.1 Modulation transfer, objective.

a. Test equipment.

(1) Collimator large enough to fully illuminate the system being tested and whose focal length is at least 10 times the effective focal length of the system being tested.

(2) Lens system holder and optical bench.

(3) Modulation transfer function (MTF) analyzer.

(4) Light source operating at a black body color temperature of 2870° K, ±100° K and power supply.

(5) X-Y recorder.

(6) Optical slit which will give approximately a 10 micron slit image at the input to the MTF analyzer.

b. Test procedure. Place the objective lens between the collimator and input optics of the MTF analyzer so their optical axes are alined with one another. Place the slit at the focal point of the collimator and illuminate it with the 2870° K light source. Visually focus the microscope objective on the MTF analyzer until the projected slit image becomes sharp. Verify that the slit image is visible over the entire face of the microscope objective. Vary the focus of the MTF analyzer for maximum high frequency response. Set the zero axis of the X-Y recorder and then adjust the MTF analyzer bias adjustment for 100 percent response at the MTF analyzer's lowest spatial frequency. (This lowest spatial frequency should be not more than 2 line pairs per millimeter when multiplied by the magnificaiton power of the microscope objective.) Verify that the zero line has not been changed. Determine the MTF from the final graph after all previous adjustments have been made. Rotate the lens assembly 90° in a plane perpendicular to its optical axis and repeat the above test. The modulation transfer should not vary more than 5 percent from the values previously obtained. Failure to conform to 3.5.7 shall constitute failure of the test.

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for effective and responsive command and control of nuclear weapons.

4.7.2 Modulation transfer, eyepiece. Perform the eyepiece lens modulation transfer test as described in 4.7.1 with the exception that a circular aperture stop of 7 millimeters, ±0.1 millimeter shall be placed between the eyepiece and the collimator at the eye relief distance centered with respect to the optical axis of the eyepiece. The eyepiece lens shall be positioned away from the photodetector and facing the collimator. Failure to conform to 3.5.8 shall constitute failure of this test.

4.7.3 Field of view. The test instrument shown in Figure 3 shall be used to project a spot of light or point source onto the objective lens of the viewer. The viewer shall then be pivoted either to the left or right until the point source seen through the viewer is at the edge of the field of view. The viewer shall then be pivoted in the opposite direction until the point source is at the opposite edge of the field of view. The total angle through which the viewer was pivoted shall be measured and the values recorded. A field of view less than 180 milliradians shall constitute failure of this test.

4.7.4 Stray light luminance. The apparatus shown in Figure 4 shall be used to measure the stray light luminance. The 2870° K tungsten light source (No. 1) employed in illuminating the white surround shall be positioned so as to provide a nearly uniform brightness of 5×10^{-3} footlambert over the surface. The 2870° K tungsten light source (No. 2) employed in illuminating the disk shall be positioned so as to provide a uniform brightness of 5×10^{-3} footlambert over the surface. The eye-corrected photodetector (see 6.3.4) shall have an aperture of 7 mm. Adjust the objective lens of the viewer for infinity focus. Compare the viewer light output with the light source (No. 1) OFF and light source (No. 1) ON. A measurement shall be made with light source (No. 2) illuminating the surface of the disk. Any change of more than 3.0 percent in light output in the measurements with light source (No. 1) OFF and the light source (No. 1) ON shall constitute failure of this test. NOTE: Care shall be taken in arranging the illuminating source (light source No. 1) so that no light falls on the surface illuminated by light source No. 2.

4.7.5 Focus adjustment.

4.7.5.1 Objective lens. The viewer shall be focused on a collimated image by adjusting the objective focus and adjusting the eyepiece diopter ring. Infinity focus shall be achieved between 20° and 120° clockwise rotation from the objective counterclockwise stop. The viewer shall also be focused on an object at a distance not greater than 4 meters by adjusting the objective focus and adjusting the eyepiece diopter ring. Using a torque wrench and adapter rotate the torque wrench first clockwise and then counter-clockwise and observe the readings. Inability for the objective lens to sharply focus a collimated image simulating infinity and an object at not more than 4 meters from the objective lens or either of the two torque readings less than 8 or greater than 26-inch pounds shall constitute failure of this test.

4.7.5.2 Eyepiece lens. Place a target approximately 1/2 inch in front of the eyepiece lens. Adjust the eyepiece to an extreme diopter setting as indicated on the eyepiece. Position a diopter tester against the eyepiece lens and adjust until the target seen through the eyepiece and diopter tester

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combination is in clear focus. Repeat the above procedure after adjusting the eyepiece lens for the other extreme diopter setting. Using a torque wrench and adapter rotate the torque wrench first clockwise and then counterclockwise and observe the readings. Failure of the eyepiece readings to coincide with the diopter tester readings within plus or minus 0.25 diopter, failure of the eyepiece to be adjustable from plus to minus 4 diopters or either of the two torque readings in excess of 7.0 inch-pounds shall constitute failure of this test.

4.7.6 Image roll. Position the viewer and target as shown in Figure 5 such that the target is perpendicular to the optical axis of the viewer. Focus the viewer on the target and adjust the alignemnt until a point on the viewer reticle is center in the target circle. Identify the starting and stopping positions on the objective lens and housing of the viewer. Slowly rotate the objective lens through 360°, keeping the target in focus. Failure of the point on the reticle to remain inside the circle on the target shall constitute failure of the test.

4.7.7 Boresight adjustment. Use the test equipment shown in Figure 3 with the target consisting of a circle and crosshair subtending 3° of the field of view. The target shall be concentric with the optical axis of the telescope. With both azimuth and elevation knobs adjusted to the center of their movement, center the reticle on the target crosshair. By scanning across the clear aperture of the eyepiece, check for any parallax between the reticle and image. Adjust the azimuth knob until the center of the reticle reaches one extremity of the 3° circle and then back to the opposite side. Return the reticle to the center of the circle. Repeat this procedure using the elevation knob. Failure of the boresight to obtain these adjustments, evidence of one adjustment affecting the other or evidence of any parallax between the reticle and crosshair shall constitute failure of this test.

4.7.8 Click movement. The test equipment described in Figures 6 and 7 shall be used to measure the boresight movement. Measurements shall be taken at both extremes and in the center of the adjustment for both elevation and azimuth. Determine the center of the azimuth adjustment and turn the azimuth adjustment to a point between 40 and 50 clicks from the center. Aline the viewer so that the reticle pattern is centered on the main crosshairs of the target (see Figure 5) turn the azimuth adjustment one click in either direction and observe the position of the crosshair relative to the two lines A and B that are applicable. Turn the azimuth adjustment two clicks in the opposite direction and again observe the position of the crosshairs relative to lines A and B this time 180° from the previous A and B lines. Repeat the test for a point within 5 clicks on the opposite side of the center. Repeat the 6 position procedure for the elevation adjustment. Using a torque wrench and adapter secure the torque wrench and adapter to the elevation knob. Rotate the torque wrench first clockwise and then counterclockwise and observe the readings. Repeat this procedure for the azimuth knob. Failure of the reticle pattern to fall between the two lines A and B for anyone of the twelve observations on the target or any of the four readings in excess of 8 inch pounds shall constitute failure of this test.

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4.8 Electromagnetic compatibility. The first article viewer equipped for the reduction of radio interference in accordance with 3.6 shall be tested as specified in MIL-STD-461. The supplier shall furnish the contracting officer the reports required by MIL-STD-461, except that the time interval for submission of reports shall be as specified (see 6.2). Upon approval of the reports by the contracting officer and provided all other requirements of the specification are met, the first article viewer shall be used as a model for all production viewers.

4.9 Inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Sight, Night Vision; Individual Served Weapons AN/PVS-2B shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Sight, Night Vision; Individual Served Weapons AN/PVS-2B shall be dried in accordance with applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Sight, Night Vision; Individual Served Weapons AN/PVS-2B. Each sight, night vision shall be individually unit packed method III as follows: place each unit within the spaces provided in the case. Close the cover and secure the fasteners. Cushion the case on all surfaces with cells or pads or both fabricated of fiberboard conforming the PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming the MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned case within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each sight, night vision shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

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5.1.3 Commercial preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Sight, Night Vision; Individual Served Weapons AN/PVS-2B, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Sight, Night Vision; Individual Served Weapons AN/PVS-2B, unit packed as specified in 5.1, shall be packed within a close-fitting fiberborad box confrming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Sight, Night Vision; Individual Served Weapons AN/PVS-2B, unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming the MIL-STD-147. The pallet

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shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming the PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The viewer is intended for use as a hand-held or individual served weapons-mounted night observation device. In addition, it can be used as a countermeasure aid in the night detection of enemy use of active systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Quantity of first article viewers to be furnished, (see 4.3).

c. Quantity and type of image intensifier assemblies to be furnished (see 3.9).

d. Time interval for submission of reports (see 4.8).

e. Level A, level B or Commercial preservation and packing (see section 5).

f. Disposition of Group C failures (see 4.5.4.2).

g. Enviromental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 Definitions. The following definitions shall apply throughout this specification.

6.3.1 Damage.

6.3.1.1 Shock and vibration damage. Includes any loosening of parts, change of sighting adjustment, failure or breakage of any hardware, connection or

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component, any change in quality of the image; any evidence of intermittent electrical contact, any change in boresight; any shift or alinement.

6.3.1.2 Operation and storage damage. Includes any loosening deterioration, corrosion of any finish, hardware, optical coating, connection or component; condensation of moisture within the telescope and high voltage breakdown; failure to operate in accordance with provisions of this specification.

6.3.2 Room temperature. Room temperature is defined as plus 23°C, ±3°C.

6.3.3 Modulation transfer function (MTF). A measure of the performance of a lens assembly as an image-producing device. In effect it is the frequency response of the assembly obtained by measuring the degradation of a slit image, as correlated to a variable frequency sine-wave pattern.

6.3.4 Eye-corrected photo-detector. For the purpose of this specification an eye-corrected photo-detector is a photo-detector or brightness meter that measures light in the particular measurement concerned within ±15 percent of that which would be obtained if the measurement were made with a device which was perfectly corrected to the Standard International Commission of Illumination (CIE) curve for photopic vision of the human eye.

6.4 Government loaned property. The contracting officer will arrange the delivery of Government loaned property specified in 3.9. The contracting officer will also arrange for the return of any unexpended rounds.

Custodians:

Army - EL
Navy - EC
Air Force - 99

Review activities:

Army - EL, MU, ME
Navy - EC
Air Force - 99

User activities:

Army - WC, MI
Navy - EC
Air Force - 99

Preparing activity:

Army - EL

Project 5855-0016

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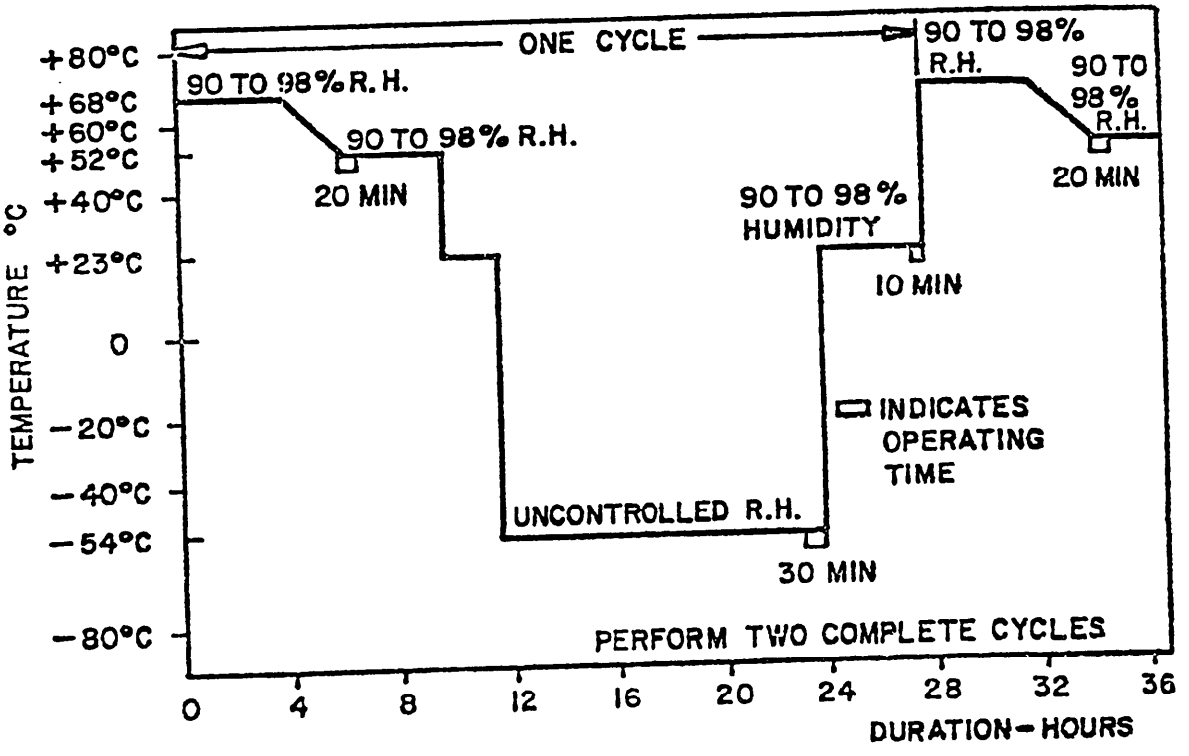


FIGURE 1
STORAGE, OPERATION AND THERMAL SHOCK

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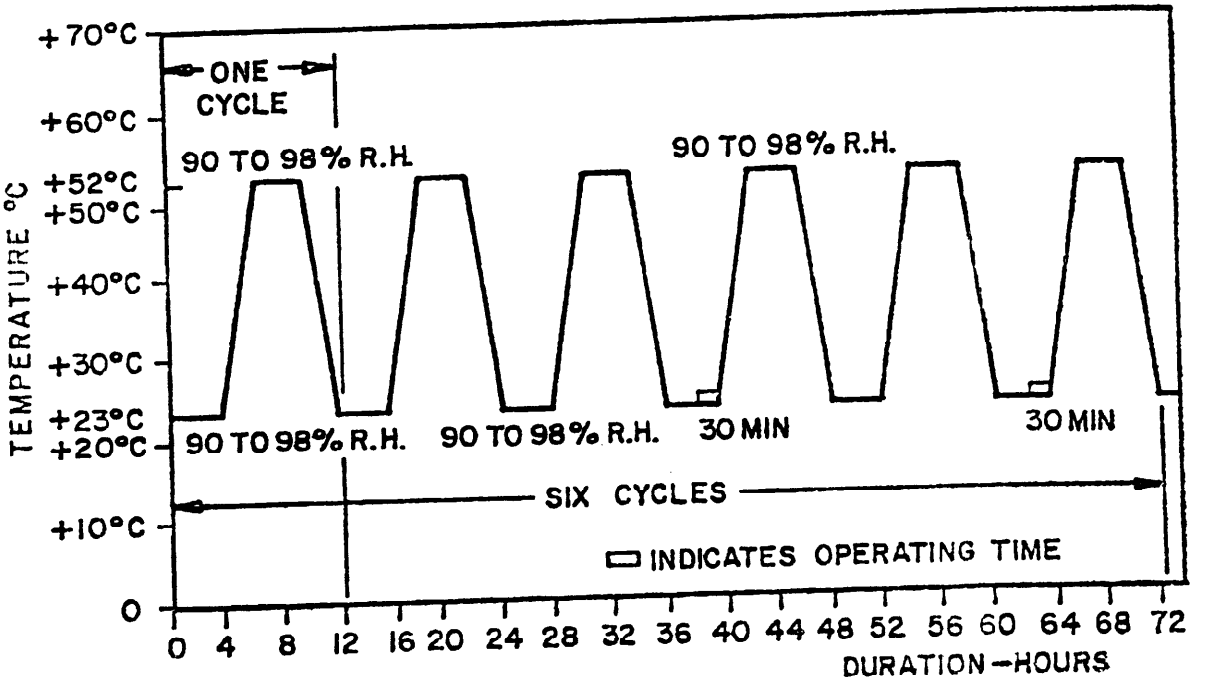


FIGURE 2

RELATIVE HUMIDITY TEST CYCLES

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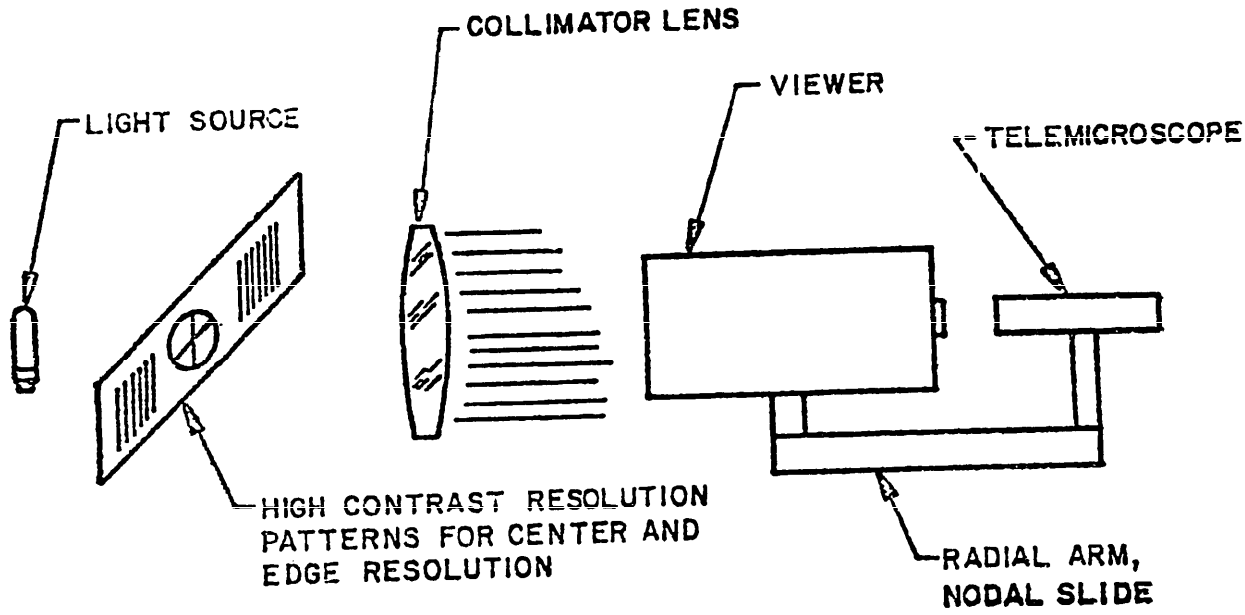
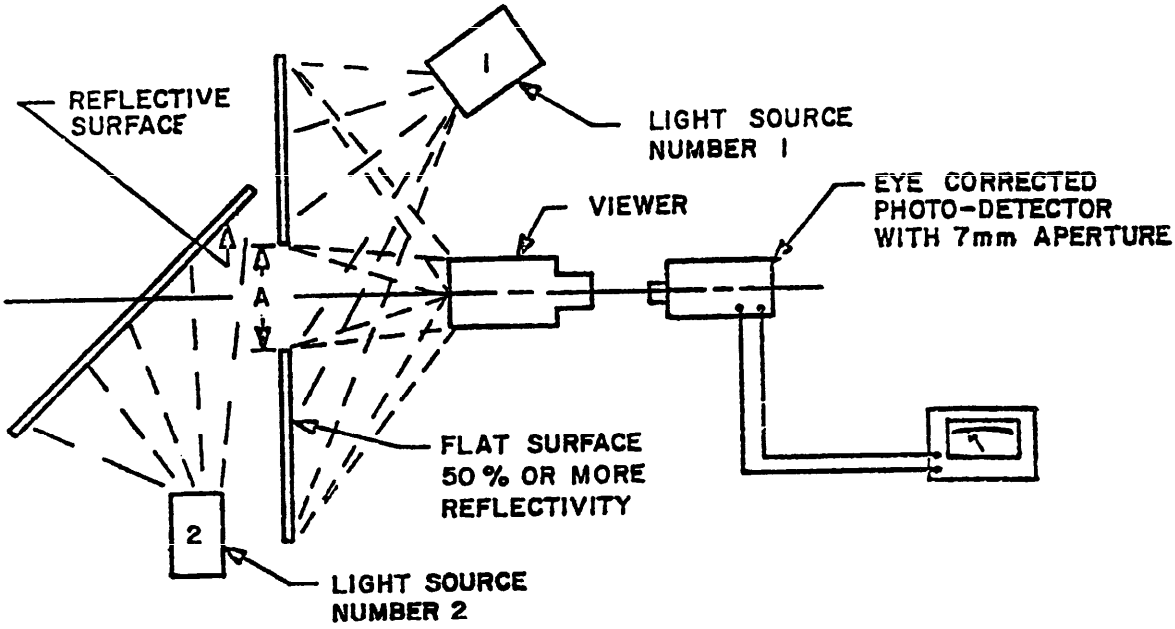


FIGURE 3

RESOLUTION, FIELD OF VIEW AND RETICLE ADJUSTMENT TESTER

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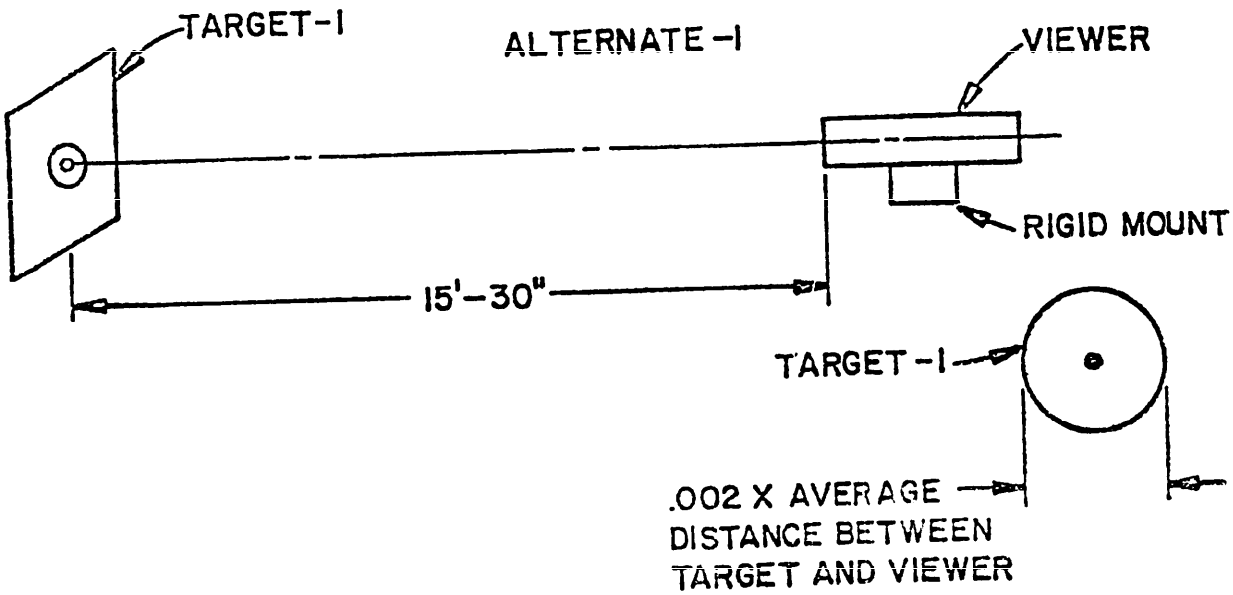
NOTES:

1. DIAMETER OF CIRCULAR HOLE "A" TO BE 3.5 INCHES LARGER THAN DIAMETER REQUIRED FOR ANGULAR SUBTENSE OF 10.5°.
2. SIZE OF REFLECTIVE SURFACE TO BE ADEQUATE TO FILL SOLID ANGLE RESULTING FROM DIAMETER NECESSARY UNDER NOTE NO.1 ABOVE.
3. ANGULAR SUBTENSE OF FLAT SURFACE TO BE 120° OR MORE.

FIGURE 4

STRAY LIGHT LUMINANCE

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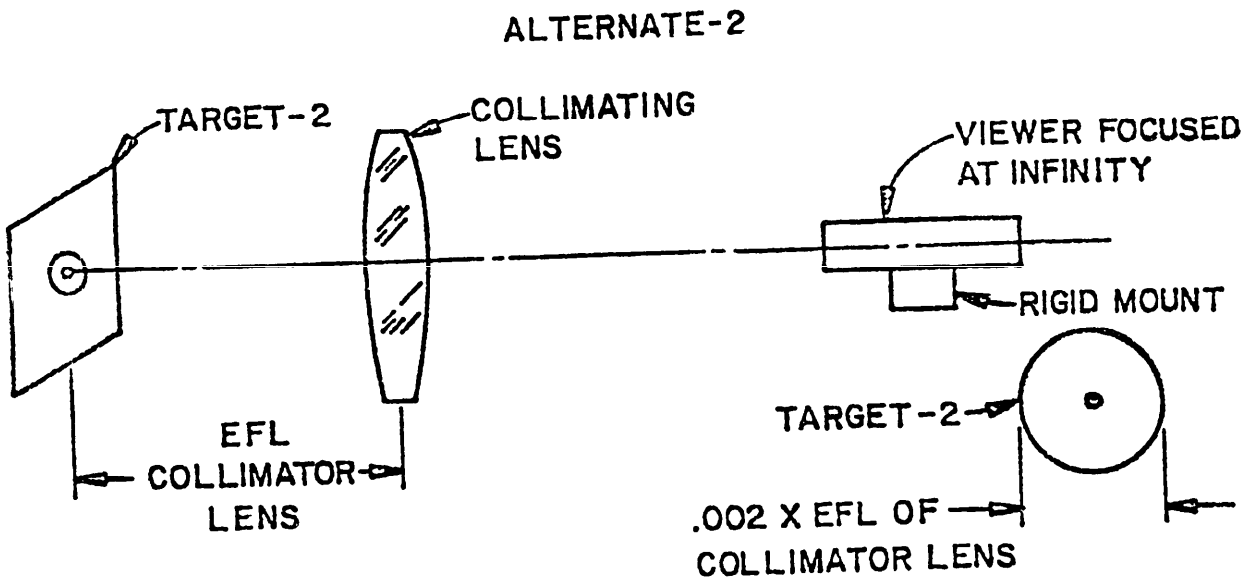


FIGURE 5

IMAGE ROLL

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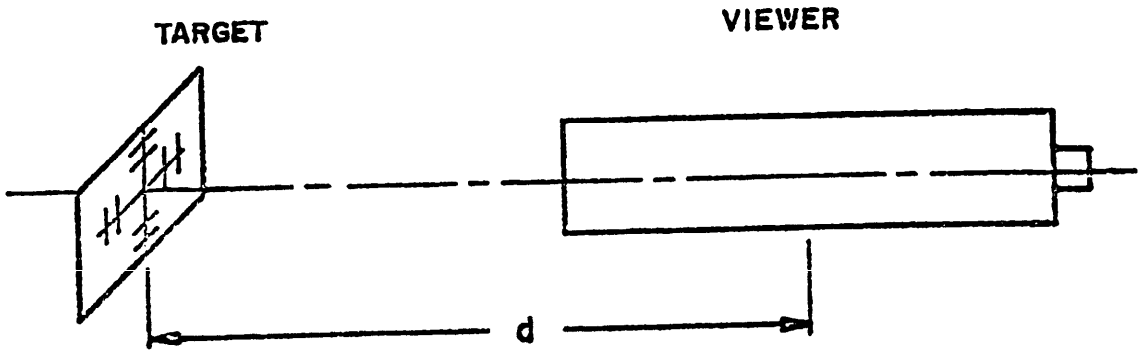
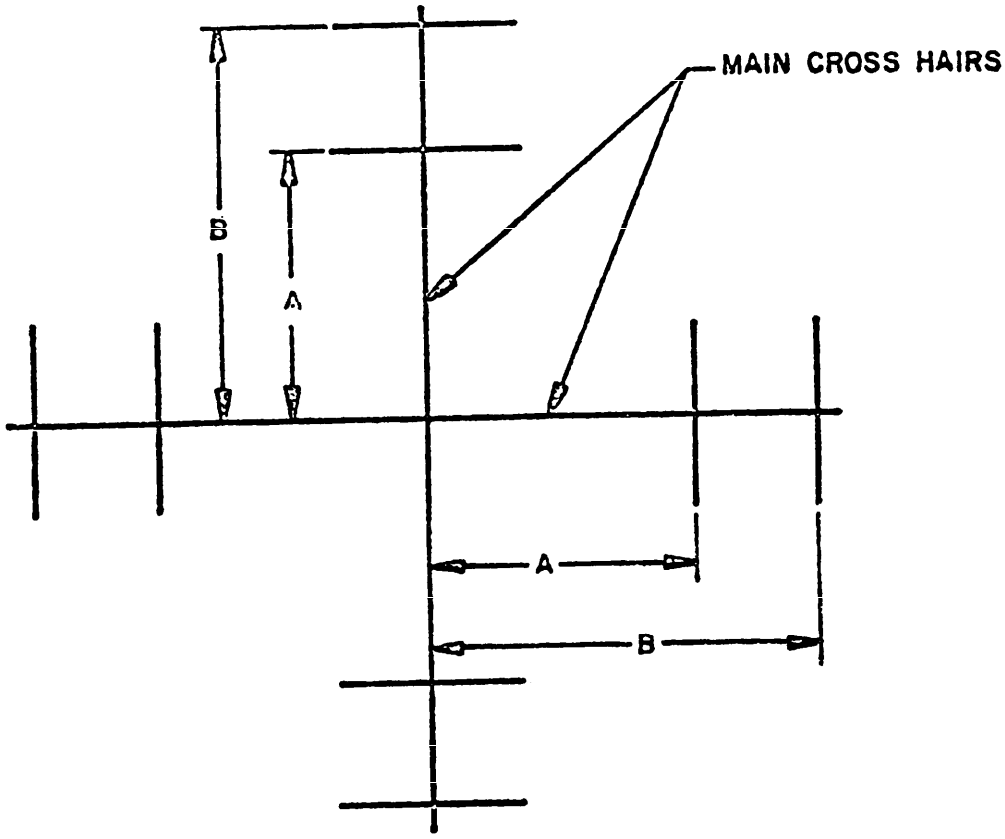


FIGURE 6

RETICLE CLICK MOVEMENT

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NOTES:

1. DIMENSION OF A EQUALS: d X .00045 INCH
2. DIMENSION OF B EQUALS: d X.00055 INCH

d= DISTANCE BETWEEN SCOPE AND TARGET IN INCHES

FIGURE 7

RETICLE CLICK MOVEMENT TARGET

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-55488B SIGHT, NIGHT VISION INDIVIDUAL SERVED WEAPONS AN/PVS-2

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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